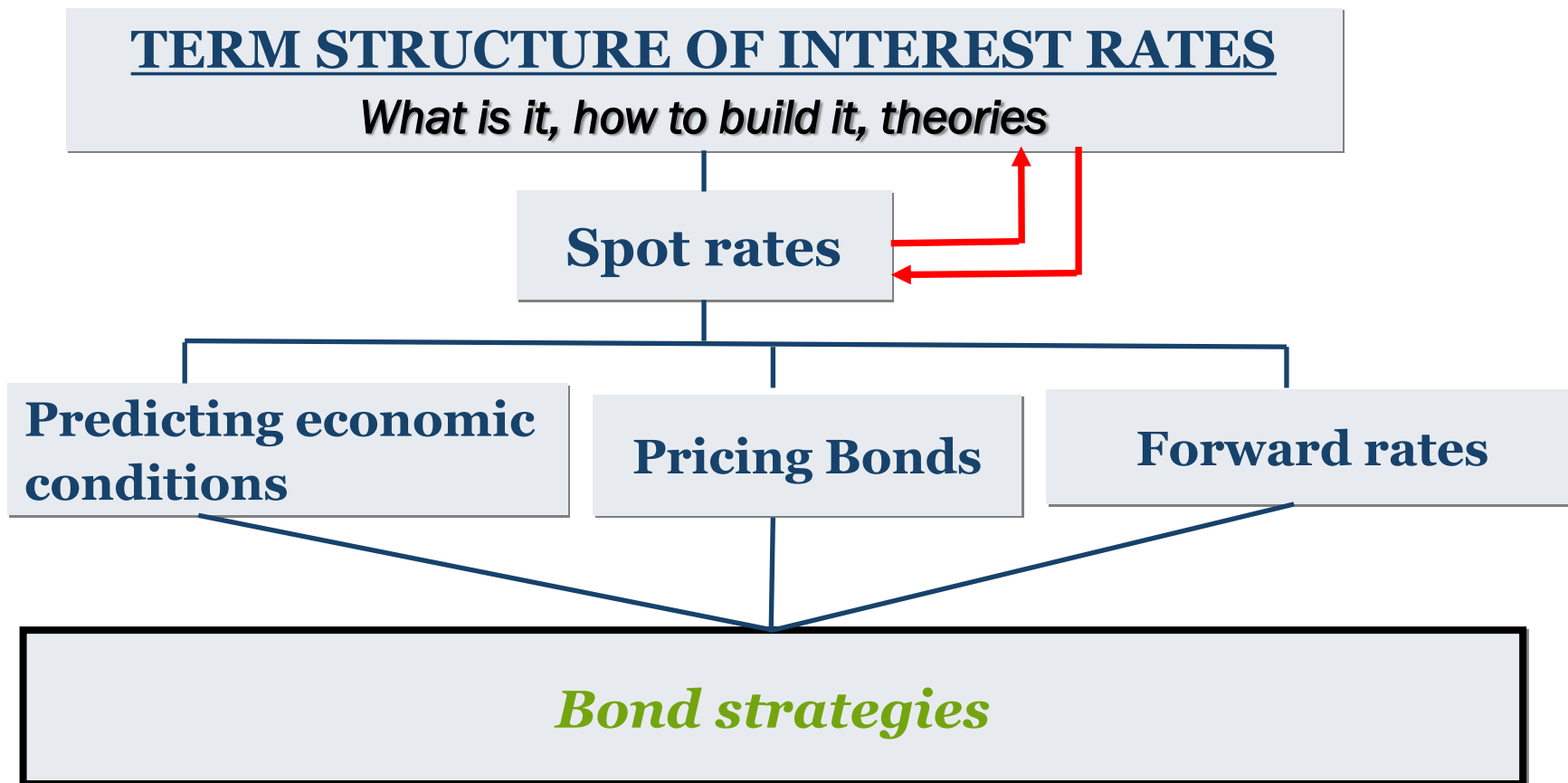


Key outcomes



❖ *The key question/key take away*

TERM STRUCTURE OF INTEREST RATES (TSI)

Bond YTM vs Time
to Maturity for a set
of Uniform bonds

YTM of Zero
Coupon Govt bond vs
their Time to Maturity

Spot rates for Govt
bonds of different
Time to Maturities

Application of spot
rates/TSI

Zero
Coupon
Govt Bonds

Plot on a
graph
Yield curve

1. Bond pricing
2. Forward rates
3. Economic conditions

*Direct
observation*

*Deriving spot
rates*

*Bootstrapping
Theoretical
spot rates*

Bond Market Flashes Recession Warning Before Round of Auctions

- Ten-year U.S. yield is below 3-mo. rate, first time since 2007
- Note sales totaling \$131 billion to test demand as yields fall

25 March 2019 at 03:00 GMT+2

Investors are about to absorb \$131 billion of Treasury note auctions at the lowest yields in months, after they piled into U.S. debt following a dovish Federal Reserve [decision](#) and fresh signs that global growth is weakening.

The week kicks off with a closely watched segment of the U.S. yield curve foreshadowing a [recession](#): The gap between 3-month and 10-year rates is now negative.

This recession indicator is flashing a warning sign

By Lucy Bayly, CNN Business

Published 9:19 AM EDT, Sat March 26, 2022

New York (CNN Business) — The bond market is flashing a warning sign that has correctly predicted almost every recession over the past 60 years a potential inversion of the US Treasury note yield curve.

Yield curve and economic conditions

Does the yield curve predict economic conditions ?

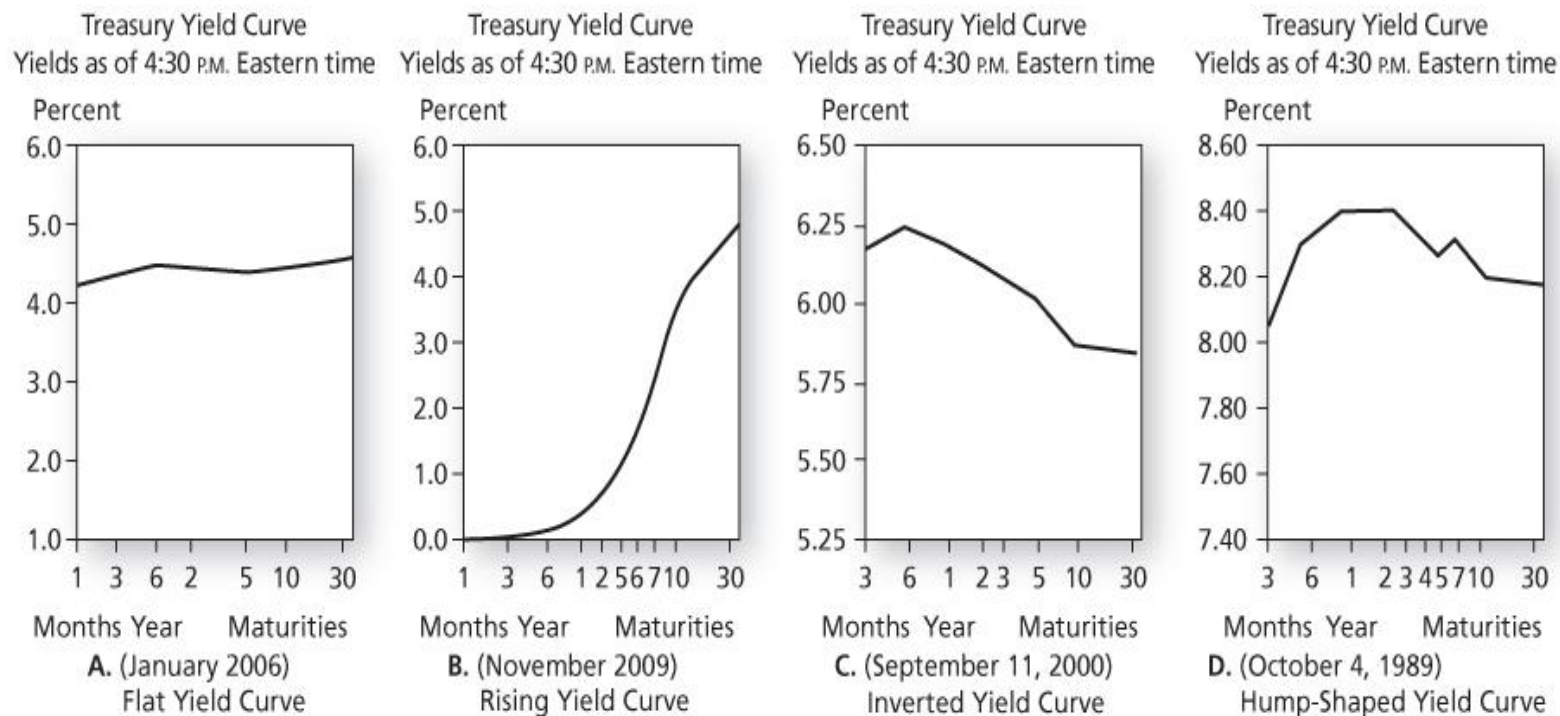


Figure 15.1 Treasury yield curves

Source: Various editions of *The Wall Street Journal*. Reprinted by permission of *The Wall Street Journal*, © 1989, 2000, 2006, and 2009 Dow Jones & Company, Inc. All Rights Reserved Worldwide.

❖ Why academics, practitioners believe the yield curve predicts economic conditions

✓ The yield curve reflect market sentiment and

✓ the market sentiment is always **"right"**

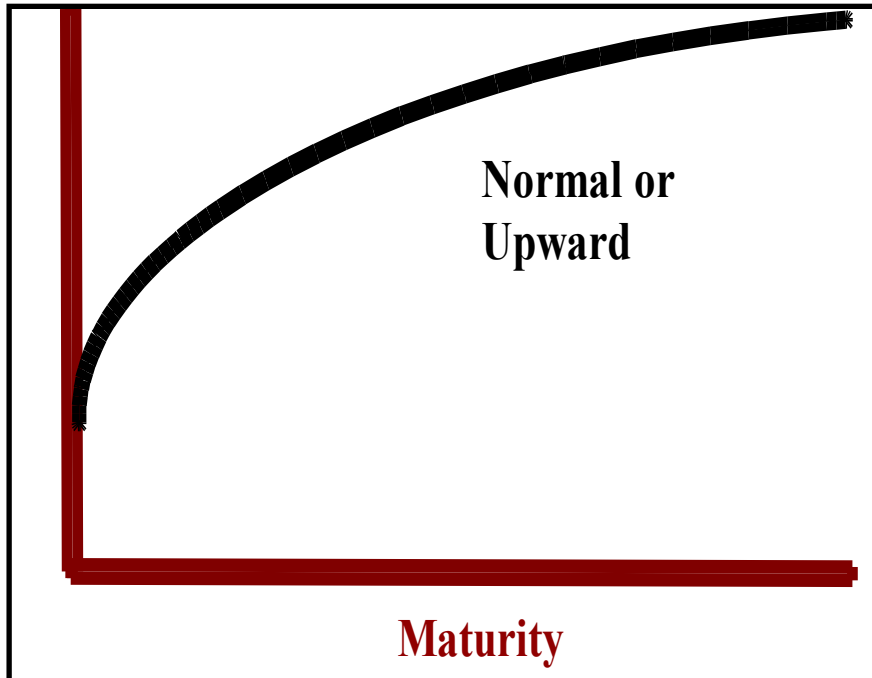
RE anticipating + positioning portfolio

because it is constructed from yields of bonds
- reflects mkt sentiment
- reflects positions traders take when anticipating certain state of

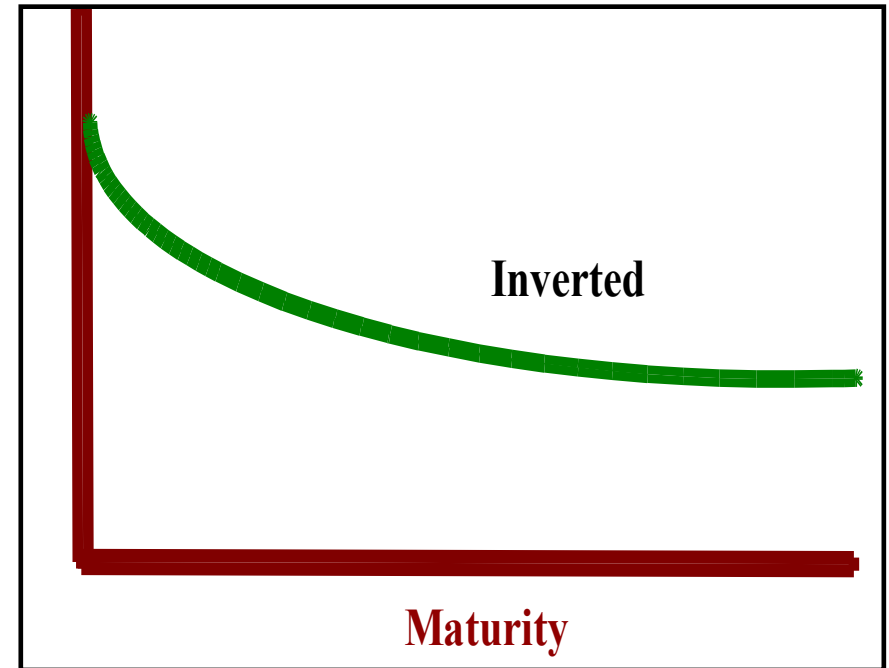
Does the yield curve predict economic conditions ?

What does the yield curve say about future economic conditions ?

boom



recession



Why is upward sloping = boom, while inverted = recession ?

NB: Upward Sloping: the steeper the curve the faster the economy is expected to grow

Yield curve and economic conditions

Why is upward sloping = boom, while inverted = recession ?

What happens in a boom or recession

Conditions in a booming economy *in general*

- ❖ high real interest rates and high inflation (high nominal rates)
 - ❖ Central bank will be hawkish *to control infl*
- because more demand for money + credit*

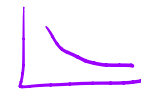
Conditions in a recession

- ❖ Low real interest and low inflation (Low nominal interest rates)
- ❖ Central bank will be dovish

NB: Nominal interest rate = Real rate + Inflation rate

Yield curve and economic conditions

Yield curve and a recession



Market anticipating a recession, eg in 1-year

[Recession=decrease in interest rates, eg from 15%-6%]

Investors/lenders shift funds from short term bonds to ^{avoid risk of lower reinvestment rate} Long term bonds ^{buy}

Borrowers/issuers will shift to short term funding-bonds

<u>Players</u>	<u>Short term bonds(ST)</u>	<u>Long term bonds(LT)</u>
Investors	Sell/Avoid ST-bonds <u>Low demand</u>	Buy LT-Bonds - to lock in high position portfolio for cap gains <u>High demand</u>
Borrowers/Issuers	Issue more ST bonds <u>High/huge supply</u>	Low issue of LT bonds <u>Low supply</u>
Effect	Prices  Yields 	Prices  Yields 
Resultant curve	Inverted Yield curve 	

1:44:40 why would investor avoid LT Bond if anticipating boom

① positioning port to buy high avoid locking low y now)

Yield curve and economic conditions

Yield curve and a boom



② Avoiding cap. losses

Market anticipating a boom, eg in 1-year

[Boom=increase in interest rates, eg from 10%-17%]

Investors/lenders shift funds from long term bonds to ^{avoid locking in low y} short term bonds

Borrowers/issuers will shift to long term funding-bonds

<u>Players</u>	<u>Short term bonds</u>	<u>Long term bonds</u>
Investors	Buy ST bonds <u>High demand</u> - positioning portfolio to buy high in future - avoiding cap. losses	Sell/Avoid LT-bonds <u>Low demand</u>
Borrowers/Issuers	Low issue ST bonds <u>Low supply</u>	More issue of LT bonds <u>High supply</u>
Effect	Prices ↑ Yields ↓	Prices ↓ Yields ↑
Resultant curve	Upward Sloping 	

Yield curve and economic conditions

The market's favorite recession indicator just flashed its starkest warning since 2007

Carmen Reinicke

Aug. 5, 2019, 03:36 PM

- The spread between three-month and 10-year Treasury yields – a relationship known as the yield curve – on Monday inverted to its widest level since 2007.
- Yield-curve inversions have long been watched for hints of an impending recession, and this is the latest signal that an economic downturn could be coming.
- This particular yield relationship has inverted before every recession over the past 50 years.
- The latest steepening of the curve was driven by escalated trade-war tensions between the US and China. *∴ correlation was a coincidence*
- Read more on Markets Insider (https://markets.businessinsider.com/?utm_source=markets&utm_medium=ingest).

Yield curve and economic conditions

The market's favorite recession indicator just flashed its starkest warning since 2007

- ❖ The spread between three-month and 10-year Treasury yields — a relationship known as the yield curve — on Monday inverted to its widest level since 2007.
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- ❖ This particular yield relationship has inverted before every recession over the past 50 years.
- ❖ The latest steepening of the curve was driven by escalated trade-war tensions between the US and China.

Yield curve and economic conditions

Case study- Recension in Argentina June 2014

Argentina economy slips into recession in first-quarter

BUENOS AIRES | BY ALEXANDRA ULMER

Argentina's economy slid into recession in the first quarter, piling fresh pressure on President Cristina Fernandez's leftist government to reach a deal with holdout creditors to avoid a new debt default.

• **Business News | Mon Jun 23, 2014 6:25pm EDT**

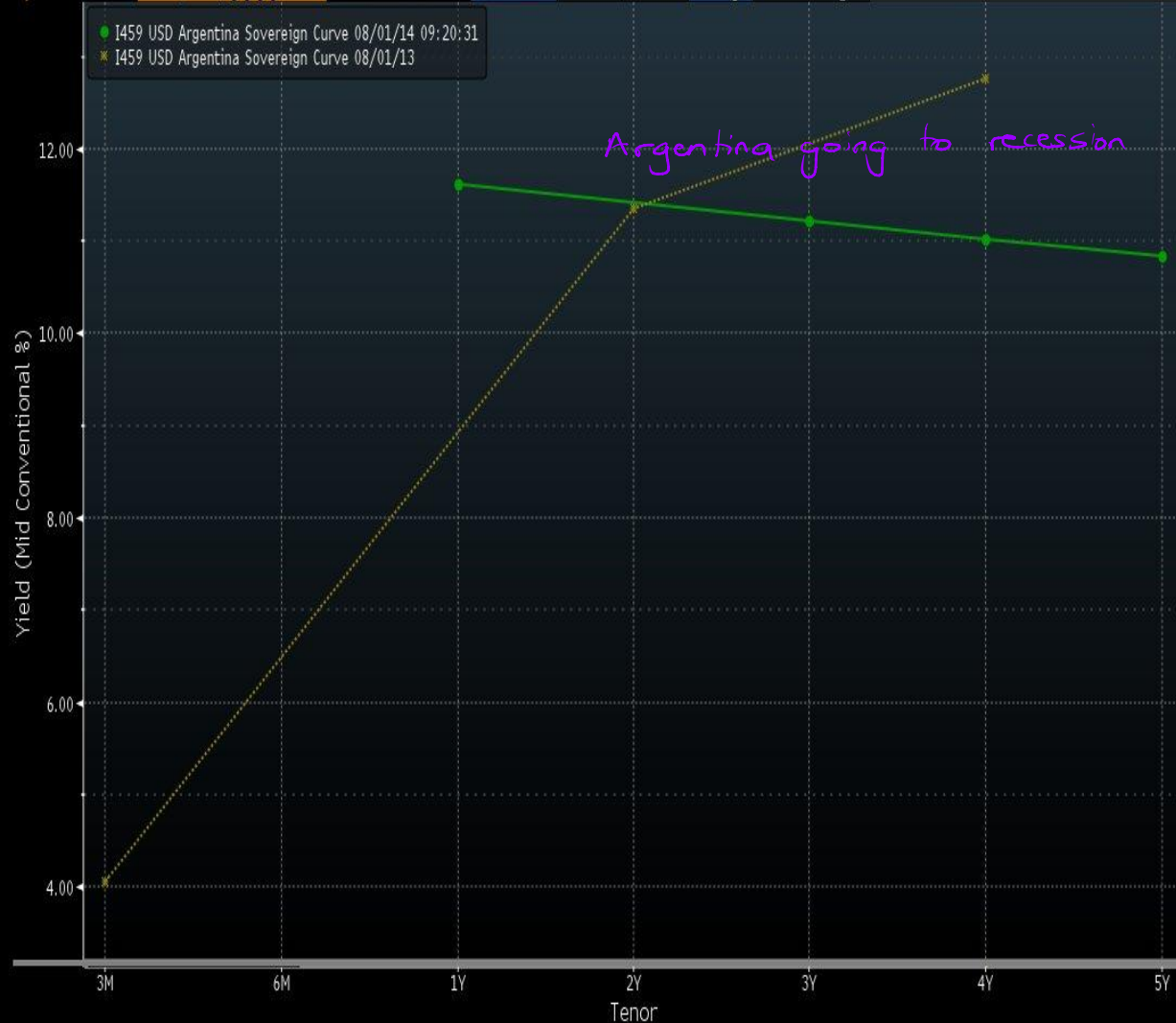
<HELP> for explanation.

Cancel: Screen not saved

USD ARGENTINA SOVEREIGN CURVE 97) Actions 98) Table Graph Curves

X-axis Tenor Y-axis Yield PCS MULT Lower Chart History Chart

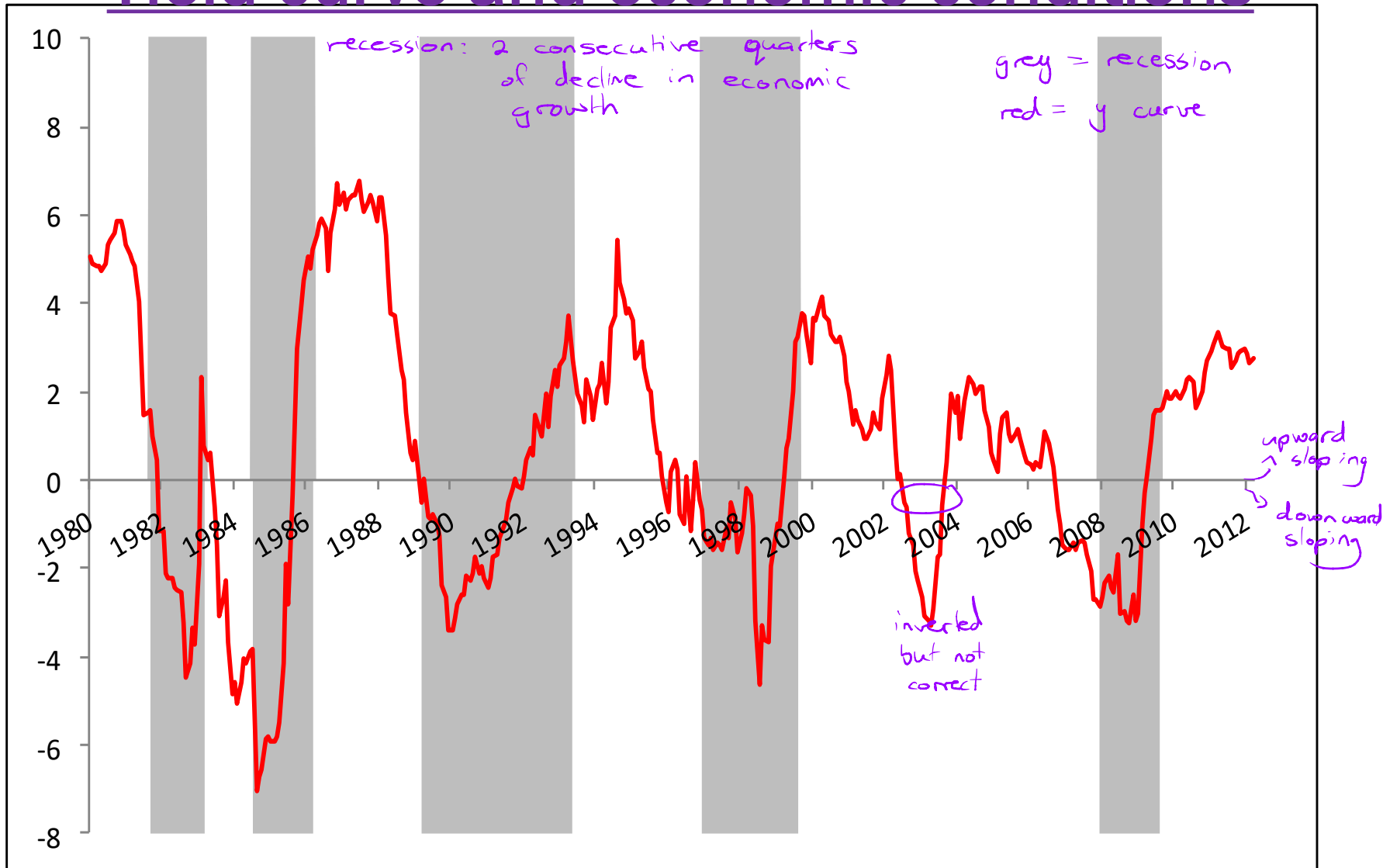
Specific mm/dd/yyyy Relative Last 1D 1W 4Q Modify Curves & Relative Value



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 - Curves related to reference security
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- Interpolate Curves

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Japan 81 3 3201 8900 Singapore 65 6212 1000 U.S. 1 212 318 2000
SN 555873 SAST GMT+2:00 H835-3228-0 01-Aug-2014 09:20:31

Yield curve and economic conditions



- ❖ Grey shaded area= recession
- ❖ Long term yields - short term yields = Positive number [Upward Sloping]
- ❖ Long term yields - short term yields = Negative number [Inverted]

Key outcomes

TERM STRUCTURE OF INTEREST RATES

What is it, how to build it, theories

Spot rates

Predicting economic conditions

Pricing Bonds

Forward rates

Bond strategies

